Jaypee Institute of Information Technology, Noida
T2 Examination, Even Semester 2026
B. Tech./ M.Tech Integrated VI Semester

Course Title: Computer Networks & Internet of Things **Maximum Time: 01 Hr**
Course Code: 18B11CS311 **Maximum Marks: 20**

Course Outcomes

CO1	Defining the basic of networking, components and underlying techniques.
CO2	Illustrate the various key protocols in OSI amdels and TCP/IP protocol suite and explain various application protocols
CO3	Examin various transpe:t protocals and its performance enhancing machenism
CO4	Determine the shortest path for the network using various routing protocols and evaluate it
CO5	Chooge IP and MAC addressing mechanisms and data link layer protocols to solve communication, error detection and correction problems.
CO6	Identification and description of various components, architectures and protocols of Internet of Things(IoT) and their real-life problemes.

Note: Attempt all questions

Q.1. Answer the following based on the given IP datagram header (in hexadecimal):

4A 10 01 F4 12 34 20 64 40 11 00 00 AC 10 01 02 AC 10 01 0A

a) Is it the first fragment? Justify.
b) How many more routers can the packet traverse before being discarded?

[CO5(Applying), 2 marks]

Q.2. A source host transmits an IPv4 datagram to a destination host with total length =5640 bytes, header=20 bytes. Identification=37921, TTL=128, and flag DF=0, MF=0, with fragment offset=0. The datagram traverses three networks with MTUs 2000 bytes, 1500 bytes, and 1800 bytes, respectively. Determine the number of fragments created at each stage (each network) and specify, for all fragments at each stage and at the final destination, their total length, fragment offset, MF flag, and data bytes ranges.

[CO5(Applying), 5 marks]

Q.3. A client-server TCP connection has MSS=1300 bytes and server receive buffer=8000 bytes. The server has received 5000 bytes, out of which 2000 bytes are processed. The Client application is currently generating data in very small increments of 100 bytes every few milliseconds.

a) Find the advertised window (rwnd).
b) If client sends data in 100-byte chunks, explain using Nagel's Algorithm whether data is sent immediately, and what happens if 1200 more bytes arrives while waiting for ACK.

[CO3(Analysing), 2 marks]

Q.4. A sender transmits 25 packets using an ARQ protocol. The sequence number field is 3 bits long. It is observed that every 6th packet (i.e. packets 6, 12, 18 and 24) is lost during transmission, while all other packets received correctly. ACKs are assumed to be error-free. Determine the total number of packet transmissions (including retransmissions) using the Go-Back-N and selective repeat protocol. Show the step-by-step transmission process at each stage.

[CO3(Analysing), 3 marks]

Q.5. Suppose a TCP Reno sender A needs to transmit a 26 KB file to receiver B. The MSS=2KB, so the file is divided into segments. The sender starts in slow start with cwnd=1 MSS and ssthresh=4MSS. After the initial sequence number exchange, the receiver advertises a receiver window(rwnd) that varies during transmission as follows: 1, 5, 2, 4, 4, 8, 3. The receiver sends an ACK for every segment, and only the latest ACK in around carries the updated rwnd. A timeout (RTO) occurs for segment 4.

which is retransmitted in the next round. Answer the following:

a) Based on the given scenario, determine and illustrate the complete evolution of the congestion window (cwnd) by drawing a cwnd vs RTT growth diagram.

b) How many transmission rounds (RTTs) occur in this communication? What will be the effective window size during last transmission round?

c) What will be the values of congestion window size and ssthresh when timeout (RTO) occurs for segment 4?

d) What is the congestion window when the 9th segment is being sent?

e) Consider a TCP connection with an initial RTO of 4 seconds. The RTT for all successful transmissions is 2 seconds. If a timeout occurs for segment 4 and TCP retransmits it in the next round, what will be the updated RTO during retransmission?

[CO3(Analysing), 2+1+1+1+3=8 marks]